MOBILITY-INFORMED AND MECHANISM-LED RENEWAL EQUATIONS

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ABSTRACT

Renewal equations have become popular models for both directly transmitted and vector-borne diseases. These equations now form the basis for most popular estimators of the instantaneous reproduction number $R(t)$ to track how an epidemic is evolving across space and time. Renewal equations have an important limitation; they assume homogeneous mixing and $R(t)$ is most often estimated for each spatial location independently. However, human movement can initiate, sustain, and prolong outbreaks. Here, we derive new mechanism-led renewal equations for both directly transmitted and vector-borne diseases. These equations directly integrate human movement to jointly estimate inward and estimate $R(t)$ for each individual locations, for pairwise combinations of locations, and across an entire network of locations. Using both simulated and empirical data, we demonstrate the bias and misleading effects of not incorporating human movement, of not accounting for within-day movement and commuting, and of using poor-quality mobility data and approximations. We further demonstrate the effects of different types and amounts of human movement by identifying which mobility features matter most. This information is then used to inform targeted intervention options in practical examples for different pathogens (SARS-CoV-2 and dengue) in different cities, where we consider homogeneous and targeted mobility reductions.

Goals:

- Show effect of not incorporating human movement (estimate R_t for each patch separately)
- Show effects of different amount and types of human movement
- Which mobility features matter most?
- Which interventions most effectively reduce overall $\mathcal{R}$ or local R_j under each mobility regime + pathogen
- **Plan:**
- Diff cites (Ho Chi Minh City + Santiago) + mobility patterns, diff pathogens (COVID + dengue), diff mobility data (mobile phone, colocation)
- Intervention options: Reduce movement homogeneously, targeted reductions on high-mobility neighbourhoods (outsized reductions), and also compare to reducing contact rate within neighbourhoods
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1 Introduction

For diseases that affect humans, human movement plays a crucial role in epidemic spread, irrespective of the route of transmission (direct, sexual, or vector-borne). Human movement enables introductions, continued establishment, persistence, spatial spread, and reintroductions **insert citations**.

In recent years, there has been an increasing volume of human movement data being available to researchers, and with this, an increasing focus of the role of human movement on epidemic dynamics

[Nouvellet et al.(2021)Nouvellet, Bhatia, Cori, Ainslie, Baguelin, Bhatt, Boonyasiri, Brazeau, Cattarino, Cooper, Coupland, Cucunuba, Buckee et al.(2020)Buckee, Balsari, Chan, Crosas, Dominici, Gasser, Grad, Grenfell, Halloran, Kraemer, Lipsitch, Metcalf, Meyers, Pe Indeed, there is an increasingly wider variety of the types of human mobility data that are commonly used by researchers, each of which have different strengths and limitations based on different research/policy questions and the local context.

Despite the essential role of humans and the growing availability of data, the relative roles of human movement are rarely linked to the most common methods for estimating real-time transmissibility (reproduction numbers $R(t)$) or epidemic speed (growth rates $r(t)$). Instead, multi-patch models (e.g. metapopulation or network models) are often used to understand how human movement impacts outbreaks in the mobility network as a whole, while simpler single-patch models (e.g. renewal-equation-based models) are used to estimate $R(t)$ and $r(t)$ for individual locations.

While multi-patch models can offer greater expressiveness, this can comes at the expense of increased complexity. This differs to traditional renewal equations for directly transmitted diseases and analogous equations for vector-borne disease, where a flexible yet very general model allows for estimating $R(t)$ with few assumptions. However, key limitations of the renewal equation include homogeneous mixing of individuals and the estimation of quantities such as $R(t)$ per-location independently. This is not just a statistically limiting factor but also potentially misleading as the local transmissibility can be over/under-estimated, human movement effects ignored, and public health policy misinformed, or at least not fully informed [Birello et al.(2024)Birello, Re Fiorentin, Wang, Colizza, and Valdano].

A recent approach by [Roy et al.(2025)Roy, Clapham, and Mishra] adapts the renewal equation to directly incorporate human movement. Other common approaches allow for a background term $m(t)$ which creates two competing processes; one that accounts for infection importations and another that accounts for infections generated locally/internally.

Here, we have a similar objective, yet take a slightly different approach, as we return to first principles and develop a spatially explicit and mobility-informed mechanistic model that is described by a set of PDEs, from which we derive the mechanism-led renewal equation, inclusive of mobility effects. Our resulting renewal equations differ to [Roy et al.(2025)Roy, Clapham, and Mishra] in four ways; i) we derive time-varying reproduction numbers using a mechanistic formula that integrates the instantaneous kernel, ii) we derive a wide taxonomy of reproduction numbers, kernels, and generation time distributions for all pairwise combinations of locations, for inward and outward transmission, and for the epidemic network as a whole, iii) we include within-day human movement dynamics to measure all infection opportunities for each individual who may spend fractions of their day in one, two, or more regions, and iv) we extend our new approach to mobility-informed renewal equations for vector-borne diseases and diseases with time-varying generation distributions.

2 Materials and methods

2.1 Renewal equations

History: Euler, Lotka, McKendrick

Recent years: [Fraser(2007)] introduced,

now basis for many popular software [Parag(2021), Abbott et al.(2020)Abbott, Hellewell, Thompson, Sherratt, Gibbs, Bosse, Munday, M Cori et al.(2013)Cori, Ferguson, Fraser, and Cauchemez]. Extended for vector-borne..

2.2 Deriving renewal equations from age-structured models

In [Mills et al.(2025)Mills, Alrefae, Hart, Kraemer, Parag, Thompson, Donnelly, and Lambert], Here, we provide a similar derivation for directly transmitted, from which a renewal equation can be derived.

2.3 Reproduction numbers, kernels, and generation times

Reproduction numbers are often to measure epidemic strength (how many infections are generated). At the start of an outbreak, the basic reproduction number.... Time-varying, $R(t)$. This can be defiend by

Some of these are new (from this paper) so will be pushed to results. Introduce here $R(t)$, $g(t, \tau)$, $K(t, \tau)$. Also, system-level

- First, different types of R:
- Who generates infections? $R_j(t)$
- How/where do infections occur? $R_{k \to j}(t)$ $R_{in}^{j}(t)$ (occurrence - is j experiencing net growth)

- Does the system grow? $\mathcal{R}(t)$
- For vector-borne, care needed, conditions affecting human transmission at time t
- Generation time distributions: How long until generate new infection in any other location g^c, how long between pairwise locations g^{cj}, and how long until next infection in location from any location g_c^{in}

2.4 Human mobility data

- Typical: Origin-destination matrix
- Lim of flow data; doesn't account for proportion of day spent in infection-opportunity-generating settings -> over-estimate infection opportunity/risk throughout day
- What we want is something that measures exposure risk....
- Two example ways; mobile phone ping data and co-location data, mobile phone ping data measures fraction of day spent in individual locations, co-location data can processed to obtained similar estimates.
- Alternatives: Show how bad gravity + radiation model approximations are

2.5 Other data

-Demographic: Population, age (if using for contact rates) -Environment

2.6 Model implementation and validation

-Stick to simulations for this

- Simulate from mobility-informed PDEs and estimate using both independent + mobility-informed PDE - IC: One infection in pop centre? Sens: One or more in multiple centres - Matrix form - Inference for follow-up: Plug in mobility data, don't think it's about having known generation time distribution... for directly transmitted: need contact rate, mobility network, - I guess it is also interesting that the single-patch GT distribution could be compared to per-location outward generation time distribution is the time between an infection in i and subsequent infection in any location in $\mathcal{N}(i)$

- How to compare? ELPD LOO, CRPS for R(t), coverage etc.

3 Results

3.1 Mobility-informed PDEs, renewal equations and reproduction numbers for directly transmitted diseases

- Derive for directly transmitted + VBD - done (see scribbles, mob_Derivations + mre_mob_derivations)
- Synthetic validation
- Repeat independently and compare to jointly, show bias from independent fits and from using flow data, synchrony,
- Mobility sensitivity index — percent change in metric (e.g. R(t), peak size, length of season) per % change in mobility. How does sensitivity depend on where you are in outbreak and where in the network?.

Suppose we have a mobility network with $\mathcal{L}$ locations. For a directly transmitted pathogen such as SARS-CoV-2, we can define a mobility-informed system of age-structured, conservation PDEs as:

$$\frac{\partial E^j}{\partial t} + \frac{\partial E^j}{\partial a_E} = -\psi(a_E)E^j(t, a_E), \tag{1}$$

where $E^j(t, a_E)$ denotes the density of individual humans in location j at time t who were exposed to a pathogen a_E time units ago and became immediately infectious. We can close the system with boundary conditions, thereby introducing mobility effects via:

$$E^j(0, a_E) = p_E^{j,0}(a_E) \tag{2}$$

$$E^j(t, 0) = \sum_{k\in\mathcal{N}(j)} \underbrace{f^{jk}(t, S) \cdot S^j(t)}_{(a)} \cdot \sum_{l\in\mathcal{N}(k)} \underbrace{f^{lk}(t, E) \int_0^t \lambda_E^l(t, a_E)E^l(t, a_E)da_E}_{(b)}, \tag{3}$$

where $p_E^{j,0}(a_E)$ denotes the initial density of infected humans, $f^{jk}(t,S)$ denotes the probability that an individual from location j is in location k at time t, $\lambda_E^k(t,a_E)$ is the transmission rate for an infected individual of infection age a_E, (a) accounts for the active population of susceptible individuals resident in neighbourhood j that are present in neighbourhood k, and (b) accounts for infected individuals that are residents in neighbourhood l transmitting to individuals present in neighbourhood k. In words, the boundary condition allows for susceptible individuals resident in location j to be infected in any location k by commuting individuals resident in any neighbourhood l.

$f^{jk}(t)$ could be processed from either the fraction of time (e.g. daily) that an individual from location j spends in location k or the probability that an individual from location j meets an individual from location j. These, $f^{jk}(t,S)$ and $f^{jk}(t,E)$, could be different for susceptible versus infected individuals and for different times. Indeed, $f^{jk}(t,E)$ could also be a function of infection age, if we suspect disease symptoms or test diagnosis to affect human movement.

$\lambda_E^{kl}(t,a_E)$ could be different for pairwise combinations of neighbourhoods and possibly a function of the activity (e.g. numbers/frequency of within-neighbourhood trips) for an individual in location k. We could have $\lambda_E^{kk}(t,a_E) > \lambda_E^{kl}(t,a_E)$ for $k \neq l$. This could reflect transmission being more likely within the home neighbourhoods due to the nature of contacts (e.g. household versus workplace). Further realism could be incorporated by not just conditioning on whether the infector is in their home residence, but also whether the putative infectee is in their home residence (i.e. the susceptible). This removes the assumption of homogeneous mixing within each location such that we have $\lambda_E^{kk}(t,a_E \mid S^k) \neq \lambda_E^{kk}(t,a_E \mid S^j)$ for $j \neq k$.

What would λ look like for SARS-CoV-2 and other directly transmitted infectious diseases? The transmission rate $\lambda_E^{kl}(t,a_E) = \frac{1}{N_{\text{eff}}^l(t)} \times \kappa^{kl}(t) \times p^k(a_E)$ is the potentially time-varying contact rate times the probability the contact results in an infection, scaled by the total population $N_{\text{eff}}^l(t) = \sum_{q\in\mathcal{N}(l)} f^{ql}(t)N^q(t)$ in location l at time t (for frequency-dependent transmission). $\kappa^{kl}(t)$ could be a function of the number of (within-neighbourhood) flows taken by the resident of location k in location l, while $p^k(a_E)$ could be based on infectiousness profiles and potentially vary based on demographics of residents of location k.

Using the method of characteristics and assuming longer-term dynamics (i.e. $t > a_E$), the boundary condition can be written as a renewal equation:

$$E^j(t,0) = \sum_{k\in\mathcal{N}(j)} \underbrace{f^{jk}(t,S)\cdot S^j(t)}_{(a)} \cdot \sum_{l\in\mathcal{N}(k)} \underbrace{f^{lk}(t,E)\int_0^t \lambda_E^l(t,a_E)\exp(-\int_0^{a_E}\psi(s)ds)E^l(t-a_E,0)da_E}_{(b)}, \quad (4)$$

In the language of [Mills et al.(2025)Mills, Alrefae, Hart, Kraemer, Parag, Thompson, Donnelly, and Lambert, Diekmann et al.(1990)Diekmann, Heesterbeek, and Metz, Diekmann and Heesterbeek(2000)], we can define instantaneous kernels at each time t that shape how infected residents of neighbourhood k of different infection ages a_E generate new infections i) in residents of neighbourhood j, $\mathbb{K}^{kj}(t,a_E)$, or ii) in residents in any neighbourhood, $\mathbb{K}_{out}^k(t,a_E)$:

$$\mathbb{K}^{kj}(t,a_E) = \sum_{l\in\mathcal{N}(k)} f^{jl}(t,S)\cdot S^j(t)\cdot f^{kl}(t,E)\lambda_E^k(t,a_E), \quad (5)$$

$$\mathbb{K}_{out}^k(t,a_E) = \sum_{j\in\mathcal{N}(k)}\sum_{l\in\mathcal{N}(k)} f^{jl}(t,S)\cdot S^j(t)\cdot f^{kl}(t,E)\lambda_E^k(t,a_E) \quad (6)$$

By integrating the instantaneous kernel over all infection ages and summing over locations, we can recover different types of instantaneous reproduction numbers:

$$R^{kj}(t) = \int_0^t \mathbb{K}^{kj}(t,a_E)da_E \quad (7)$$

$$R_{out}^k(t) = \sum_{j\in\mathcal{N}(k)} R^{kj}(t) = \sum_{j\in\mathcal{N}(k)}\int_0^t \mathbb{K}^{kj}(t,a_E)da_E \quad (8)$$

where the pairwise $R^{kj}(t)$ accounts for the expected number of new human infections generated by individuals from neighbourhood k in j, while the row sums $R_{out}^k(t)$ of matrix $\mathbf{R}(\mathbf{t})$ below are the outward reproduction numbers defining the expected number of new human infections generated by neighbourhood k in any neighbourhood. The former is instructive for estimating the pairwise transmissibillity patterns necessary for between- and within neighbourhood transmission, while the latter is useful for estimating whether a given neighbourhood is a net source of transmission.

We can assemble all our spatially explicit reproduction numbers into a matrix form:

$$\mathbf{R}(\mathbf{t}) = \begin{pmatrix} R^{11}(t) & R^{12}(t) & \dots & R^{1N}(t) \\ R^{21}(t) & \dots & \dots & R^{2N}(t) \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & R^{NN}(t) \end{pmatrix},$$

from which the system-/network-level reproduction number can be derived as the spectral radius of this reproduction matrix:

$$\mathcal{R}(t) = \rho(\mathbf{R}(\mathbf{t})), \tag{9}$$

which measures instantaneous growth/decay of the network. This can be used to evaluate how interventions impact the epidemic as a whole (i.e. across the network). Also, the subdominant eigenvalue $\lambda_2(t)$ captures how quickly the epidemic converges to the asymptotic spatial distribution. The ratio $s(t) = \frac{|\lambda_2(t)|}{\rho(\mathbf{R}(t))}$ measures the "mixing time" of the epidemic on the network — how many generations it takes for the spatial pattern of infections to stabilise. $s(t)$ values close to 1 indicates that the epidemic maintains its current spatial configuration. [**?**] showed how this quantity governs the bias from estimating the aggregate $\mathcal{R}(t)$ for spatially structured populations without a spatially structured approach.

If this ratio is close to 1, the epidemic retains memory of its initial spatial configuration for a long time, meaning that where you seed the epidemic matters a lot for the trajectory. If the ratio is small, the epidemic rapidly "forgets" where it started and the asymptotic spatial distribution dominates early. Also, to capture the heterogeneity in outward transmission, we can compute the coefficient of variation (CV) of the row sums of $\mathbf{R}(t)$.

Relatedly, system-level growth rate:

As usual, we can derive densities for the generation time distributions by dividing the instantaneous kernel by the corresponding reproduction number, yet here we have several different types of generation time distributions:

$$g^{kj}(t, a_E) = \frac{\mathbb{K}^{kj}(t, a_E)}{\int_0^t \mathbb{K}^{kj}(t, a_E) da_E} = \frac{\mathbb{K}^{kj}(t, a_E)}{R^{kj}(t)} \tag{10}$$

$$g^k_{out}(t, a_E) = \frac{\sum_{j \in \mathcal{N}(k)} \mathbb{K}^{kj}(t, a_E)}{\sum_{j \in \mathcal{N}(k)} \int_0^t \mathbb{K}^{kj}(t, a_E) da_E} = \frac{\mathbb{K}^k_{out}(t, a_E)}{R^k_{out}(t)} \tag{11}$$

where $g^{kj}(t, a_E)$ defines, at time t, the relative contribution to the number of new human infections generated in residents of location k by humans resident in location k that were infected a_E time units ago. $g^k_{out}(t, a_E)$ defines, at time t, the relative contribution to the number of new human infections generated in any location by humans resident in location k that were infected a_E time units ago.

Then, we can define our renewal equation per location, capturing how previously infected individuals generate new infections in location j:

$$E^j(t, 0) = \sum_{k \in \mathcal{N}(j)} \int_0^t \mathbb{K}^{kj}(t, a_E) E^k(t - a_E, 0) da_E \tag{12}$$

$$= \sum_{k \in \mathcal{N}(j)} R^{kj}(t) \int_0^t g^{kj}(t, a_E) E^k(t - a_E, 0) da_E. \tag{13}$$

If we set $f^{jk}(t, S) = \delta_{jk}$ and $f^{jk}(t, E) = \delta_{jk}$ for all locations j, then we recover the familiar single-patch renewal equation:

$$E^j(t, 0) = \int_0^t \mathbb{K}^{jj}(t, a_E) E^j(t - a_E, 0) da_E \tag{14}$$

$$= R^{jj}(t) \int_0^t g^{jj}(t, a_E) E^j(t - a_E, 0) da_E. \tag{15}$$

Alternatively, we have between-location renewal equations

$$E^{k \to j}(t, 0) = R^{kj}(t) \int_0^t g^{kj}(t, a_E) \cdot E^c(t - -a_E, 0) da_E, \tag{16}$$

which may be useful to capture how infections are propogating across the network at any given time.

We can therefore set up a renewal equation matrix $\mathbf{E}(t, 0)$ with individual elements $\mathbf{E}(t, 0)_{jk}$ describing the number of new infections generated by previously infected individuals from neighbourhood k in neighbourhood j. Rows thereby sum to the number of new infections occurring in neighbourhood j at time t.

$$\mathbf{E}(\mathbf{t}, \mathbf{0}) = \begin{pmatrix} E^{1\to 1}(t) & E^{1\to 2}(t) & \dots & E^{1\to N}(t) \\ E^{2\to 1}(t) & E^{2\to 2}(t) & \dots & \dots \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & E_{NN}(t) \end{pmatrix}$$

The diagonal elements describe the within-neighbourhood transmission from the renewal equation (akin to the traditional single-location equation), while other elements describe outside-neighbourhood transmission.

To capture the inward transmissibility pressure, we can define the expected number of new infections occurring in residents of location j at time t by infected individuals resident in any location as:

$$R_{in}^{j}(t) = \sum_{k\in\mathcal{N}(j)} R^{kj}(t) = \sum_{k\in\mathcal{N}(j)} \int_0^t \mathbb{K}^{kj}(t, a_E) da_E, \tag{17}$$

where the usual threshold property, $R_{in}^{j}(t)$ above/below 1, retains familiar meaning from the univariate setting for each location j. Likewise, we can define the inward generation time distributions and instantaneous kernels per location:

$$\mathbb{K}_{in}^{j}(t, a_E) = \sum_{k\in\mathcal{N}(j)} \mathbb{K}^{kj}(t, a_E), \tag{18}$$

$$g_{in}^{j}(t, a_E) = \frac{\mathbb{K}_{in}^{j}(t, a_E)}{R_{in}^{j}(t)}, \tag{19}$$

which capture respectively, at time t, how many expected new infections and the relative contribution to new infections by individuals of infection age a_E resident in any location.

System-level GT, kernel

Epidemic speed — the spatial spreading rate; i.e. the average spatial generation distance — how far, on average, a secondary infection is from its primary infection. Combined with the mean generation time, the ratio gives you an estimate of the epidemic's spatial velocity: $v(t) = \frac{\bar{d(t)}}{g(t)}$

Adding an incubation period is rather simple, but does require an extra state variable I. Repeating procedure...

3.2 Mobility-informed PDEs, renewal equations and reproduction numbers for vector-borne diseases

Densities for

$$\frac{\partial E^j}{\partial t} + \frac{\partial E^j}{\partial a_E} = -b_{EI}^{j}(t, a_E)\, E^j(t, a_E), \tag{20}$$

$$\frac{\partial I^j}{\partial t} + \frac{\partial I^j}{\partial a_I} = 0, \tag{21}$$

$$\frac{\partial W^j}{\partial t} + \frac{\partial W^j}{\partial a_W} = -b_{WV}^{j}(t, a_W)\, W^j(t, a_W) - \mu^j(t) W^j(t, a_W), \tag{22}$$

$$\frac{\partial V^j}{\partial t} + \frac{\partial V^j}{\partial a_V} = 0, \tag{23}$$

where E and I denote infected and infectious humans respectively and W and V denote infected and infectious mosquitoes respectively. We close the system with the following boundary conditions.

$$E^j(t,0) = \sum_{k\in\mathcal{N}(j)} \int_0^\infty b_{VE}^{kj}\left(t,a_V\right) V^j\left(t,a_V\right) da_V, \tag{24}$$

$$= \sum_{k\in\mathcal{N}(j)} \int_0^\infty b^k(t) f^{jk}(t,S) \frac{S^{H,j}(t)}{\sum_{l\in\mathcal{N}(k)} f^{lk}(t,S) N^{H,l}(t)} V^k\left(t,a_V\right) da_V, \tag{25}$$

$$E^j\left(0,a_E\right) = p_E^j\left(a_E\right), \tag{26}$$

$$I^j(t,0) = \int_0^\infty b_{EI}^j\left(t,a_E\right) E^j\left(t,a_E\right) da_E, \tag{27}$$

$$I^j\left(0,a_I\right) = p_I^j\left(a_I\right), \tag{28}$$

$$W^j(t,0) = \sum_{k\in\mathcal{N}(j)} \int_0^\infty b_{IW}^{kj}\left(t,a_I\right) I^k\left(t,a_I\right) da_I, \tag{29}$$

$$= \sum_{k\in\mathcal{N}(j)} \int_0^\infty \frac{b^k(t) S^{M,k}(t)}{\sum_{l\in\mathcal{N}(k)} f^{lk}(t,I) N^{H,l}(t)} f^{kj}(t,I) I^k\left(t,a_I\right) p_{WI}(a_I) da_I \tag{30}$$

$$W^j\left(0,a_W\right) = p_W^j\left(a_W\right), \tag{31}$$

$$V^j(t,0) = \int_0^\infty b_{WV}^j\left(t,a_W\right) W^j\left(t,a_W\right) da_W, \tag{32}$$

$$V^j\left(0,a_V\right) = p_V^j\left(a_V\right). \tag{33}$$

where $b_{\text{VE}}^{kj}(t,a_V)$ and $b_{\text{IW}}^{kj}(t,a_I)$ represent the infection processes by which a susceptible human and mosquito resident in neighbourhood j respectively are infected by an infectious human and mosquito respectively. The development processes $b_{\text{EI}}^{j}(t,a_I)$ and $b_{\text{WV}}^{j}(t,a_V)$ represent the rate at which infected humans and mosquitoes become infectious. $p_E^j(a_E), p_I^j(a_I), p_W^j(a_W)$, and $p_V^j(a_V)$ are the initial densities for the exposed human, infectious human, exposed mosquito, and infectious mosquito populations respectively. We have assumed that neighbourhoods are sufficiently large such that mosquitoes can only infect and be infected within their home neighbourhood. This is the case for most real-world applications to surveillance data, but to account for beyond-neighbourhood movement and infection of susceptible mosquitoes, this would simply require an analogous term to $f^{jk}(t,S)$ for susceptible mosquitoes that represents the probability that the susceptible mosquito from neighbourhood j is in neighbourhood k at any time t and two additional terms to account for additional possible meeting locations with susceptible and infected humans.

$b_{VE}^{kj}\left(t,a_V\right)$ per-capita rate at which mosquito in neighbourhood k of age a_V at time t infects susceptible human from neighbourhood j.

$$\begin{aligned}
b_{EI}^j\left(t,a_E\right) &= \lambda_{E,\text{Latent}}^j(a_E)\\
b_{IW}^j\left(t,a_I\right) &= \frac{b(t)}{N^H(t)} \cdot S^M(t) \cdot p_{WI}(a_I) = \frac{b(t)}{N^H(t)} \cdot \left(N^M(t) - \int_0^\infty W(t,a_W)da_W - \int_0^\infty V(t,a_V)da_V\right) \cdot p_{WI}(a_I)\\
b_{WV}\left(t,a_W\right) &= \lambda_{\text{W, Latent}}^j(t,a_W)\\
b_{VE}^j\left(t,a_V\right) &= b(t) \cdot \frac{S^H(t)}{N^H(t)} \cdot p_{EV}(a_V) = b(t)\left(N^H(t) - \int_0^t E(t,a_E)da_E - \int_0^t I(t,a_I)da_I\right)/N^H(t) \cdot p_{EV}(a_V)
\end{aligned} \tag{34}$$

where $\lambda_{\text{Latent}}^E(a_E)$ and $\lambda_{\text{Latent}}^W(a_W)$ denote the rates at which latent periods end for infected humans and mosquitoes respectively, $N^H(t)$ and $N^M(t)$ denote the total human and mosquito populations, $S^H(t)$ and $S^M(t)$ denote the susceptible human and mosquito population counts, p_{WI} and p_{EV} denote the probability of mosquito and human infection upon infectious bite respectively, and $b(t)$ denotes the rate at which a mosquito bites a human.

Hazard rates for the human latent periods $b_{\text{EI}}^j(t,a_E)$ may differ by population demographics (e.g. older populations), while the mosquito latent peridod $b_{\text{WV}}^j(t,a_E)$ may differ by location/space (e.g. urban landscape and mosqutio adaptation).

The renewal equation for new human infections in neighbourhood j can be written as:

$$\begin{aligned}
E^j(t,0) &= \sum_{k\in\mathcal{N}(j)} \int_0^\infty b^{kj}_{VE}(t,a_V)\, V^j(t,a_V)\, da_V, \\
&= \sum_{k\in\mathcal{N}(j)} \int_0^\infty b^{kj}_{VE}(t,a_V)\, V^j(t-a_V,0)\, e^{-\mu a_V}\, da_V, \\
&= \sum_{k\in\mathcal{N}(j)} \int_0^\infty b^{kj}_{VE}(t,a_V)\, e^{-\mu a_V} \int_0^\infty b^{j}_{WV}(t,a_W)\, W^j(t-a_V,a_W)\, da_W da_V, \\
&= \sum_{k\in\mathcal{N}(j)} \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} b^k(t) f^{jk}(t) \frac{S^{H,j}(t)}{\sum_{l\in\mathcal{N}(k)} f^{lk}(t) N^{H,k}(t)} p_{EV}(a_V) b^k_{WV}(t-a_V,a_W) \\
&\quad \cdot \sum_{c\in\mathcal{N}(k)} \frac{b^c(t-a_V-a_W) S^{M,c}(t-a_V-a_W)}{\sum_{d\in\mathcal{N}(c)} f^{dc}(t-a_V-a_W) N^{H,d}(t-a_V-a_W)} f^{kj}(t-a_V-a_W) p_{WI}(a_I) \\
&\quad\quad \cdot b^c_{EI}(t-a_V-a_W-a_I,a_E)\, e^{\int_0^{a_V} -\mu^c(t-a_V-a_W+r)dr} e^{-\int_0^{a_W} b^c_{WV}(t-a_V-a_W+r,r)-\mu^c(t-a_V-a_W+r)dr} \\
&\quad\quad\quad \cdot E^c(t-a_V-a_W-a_I-a_E,0) e^{-\int_0^{a_E} b^c_{EI}(t-a_V-a_W-a_I-a_E+s,s)ds} da_E da_I da_W da_V
\end{aligned} \tag{35}$$

or compactly:

$$E^j(t,0) = \sum_{k\in\mathcal{N}(j)} \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} \sum_{c\in\mathcal{N}(k)} \mathbb{K}^{ckj}(t,a_E,a_I,a_W,a_V) \cdot E^c(t-a_V-a_W-a_I-a_E,0) da_E da_I da_W da_V \tag{36}$$

$\mathbb{K}^{ckj}(t,a_E,a_W,a_I,a_V)$ is the rate at which previously infected individuals from neighbourhood c produce secondary infections that occur in neighbourhood k for residents of neighbourhood j. This instantaneous rate is for the expected number of secondary infections occurring in residents from neighbourhood j at calendar time t from infectious mosquitoes resident in neighbourhood k, which were infected at calendar time $t - a_V - a_W$ in neighbourhood k by an infectious human resident in neighbourhood c who was originally infected at time $t - a_V - a_W - a_I - a_E$. The ordering of notation (c, k, j) reflects the resident locations of the ordered infection events (primary human infection of resident of neighbourhood c, infection of mosquito resident of neighbourhood k, and secondary human infection of resident of neighbourhood j). As in [Mills et al.(2025)Mills, Alrefae, Hart, Kraemer, Parag, Thompson, Donnelly, and Lambert], the ages in $\mathbb{K}^{ckj}(t,a_E,a_W,a_I,a_V)$ reflect the ages of the individual humans/mosquitoes at the times when they play their role in the multi-stage transmission cycle.

Note for vector-borne diseases, an extra sum appears and the next-generation operator/matrix (for transmission between locations) becomes a three-dimensional next-generation array. Why? This additional dimension reflects the additional location opportunities/possibilities for the multi-stage infection process. The originally infected human resident in neighbourhood c can generate a new mosquito infection in a neighbourhood k before a resident from neighbourhood j comes into contact with that infected mosquito from neighbourhood k.

As before, different types of human-to-human instantaneous reproduction numbers can be formed by integrating over all ages and summing over meeting/infection locations. These will retain the familiar meaning as traditional reproduction numbers. We define the pairwise $R^{cj}(t)$, the expected number of secondary human infections in neighbourhood j generated by a previously infected human resident in neighbourhood c throughout their infected lifetime if conditions affecting this human transmission route were to remain the same at time t:

$$R^{cj}(t) = \sum_{k\in\mathcal{N}(j)} \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} \mathbb{K}^{ckj}(t,a_E,a_I,a_W,a_V) \tag{37}$$

This expression sums over all locations k where the susceptible human resident in neighbourhood j can be infected by an infectious mosquito in any neighbourhood k who was themselves infected by the infected human resident in neighbourhood c.

Likewise, the outward reproduction number $R^c_{out}(t)$ is defined as:

$$R^c_{out}(t) = \sum_{j\in\mathcal{N}(c)} \sum_{k\in\mathcal{N}(j)} \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} \mathbb{K}^{ckj}(t,a_E,a_I,a_W,a_V) da_E da_I da_W da_V \tag{38}$$

This sums over all locations j where a secondary infected human could be resident to capture, at time t, how many expected infections are generated by the infected individual resident in location c, if conditions affecting human transmission were to remain the same. Note we have also summed over all possible locations where the intermediate mosquito was infected, or equivalently where the infected human resident in location j was infected. This captures whether

We can define the expected number of new infections occurring in residents of location j by infected individuals resident in any location throughout their infected lifetime if transmission conditions were to remain the same.

$$R^j_{in}(t) = \sum_{k\in\mathcal{N}(j)} \sum_{c\in\mathcal{N}(k)} \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} \mathbb{K}^{ckj}(t, a_E, a_I, a_W, a_V) da_E da_I da_W da_V, \tag{39}$$

which indicates whether a location is experiencing net instantaneous growth/decay, and whether it is a sink of transmission.

Also, we can write instantaneous kernels between two locations (rather than triplets) by summing over the locations of the mosquito resident location:

$$\mathbb{K}^{cj}(t, a_E, a_I, a_W, a_V) = \sum_{k\in\mathcal{N}(j)} \mathbb{K}^{ckj}(t, a_E, a_I, a_W, a_V) da_E da_I da_W da_V \tag{40}$$

and per-location outward instantaneous kernels

$$\mathbb{K}^c_{out}(t, a_E, a_I, a_W, a_V) = \sum_{j\in\mathcal{N}(c)} \sum_{k\in\mathcal{N}(j)} \mathbb{K}^{ckj}(t, a_E, a_I, a_W, a_V) da_E da_I da_W da_V \tag{41}$$

This suggests between- and per-location generation time distributions can also be formed as:

$$g^{cj}(t, a_E, a_I, a_W, a_V) = \frac{\mathbb{K}^{cj}(t, a_E, a_I, a_W, a_V)}{R^{cj}(t)}, \tag{42}$$

which describes, at each time t, the relative contributions of different stage-specific ages to the expected number of new human infections for residents of neighbourhood j arising from the original human infection from a resident of neighbourhood c. Again, ages evaluated at the calendar times when each stage-specific individual played their role in the transmission cycle.

$$g^c_{out}(t, a_E, a_I, a_W, a_V) = \frac{\mathbb{K}^c_{out}(t, a_E, a_I, a_W, a_V)}{R^c_{out}(t)}, \tag{43}$$

which describes, at each time t, the relative contributions of different stage-specific ages to the expected number of new human infections arising from the original human infection from a resident of neighbourhood c.

The different formulations of R, g, and K all count different classes of infections and therefore define differently the expected numbers of new infections, the times between infections, and the expected contributions of different ages to new ages.

Our renewal equation can be written as:

$$E^j(t,0) = \sum_{c\in\mathcal{N}(k)} R^{cj}(t) \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} g^{cj}(t, a_E, a_I, a_W, a_V) \cdot E^c(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W \tag{44}$$

We can also compute between-location renewal equations

$$E^{c\to j}(t,0) = R^{cj}(t) \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} g^{cj}(t, a_E, a_I, a_W, a_V) \cdot E^c(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W da_V \tag{45}$$

As before, we can set up renewal equation matrix $\mathbf{E}(t,0)$ with individual elements $\mathbf{E}(t,0)_{jc} = E^{c\to j}(t,0)$ describing the number of new human infections generated by previously infected individuals from neighbourhood c in neighbourhood c.

In matrix form,

$$\mathbf{E}(\mathbf{t},\mathbf{0}) = \begin{pmatrix} R^{11}(t) \int\int\int\int g^{11}(t, a_E, a_I, a_W, a_V) \cdot E^1(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W da_V & E^{2\to 1}(t,0) & \dots & E^{N\to 1}(t,0) \\ R^{12}(t) \int\int\int\int g^{12}(t, a_E, a_I, a_W, a_V) \cdot E^1(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W da_V & E^{1\to 2}(t,0) & \dots & E^{N\to 2}(t,0) \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & E^{N\to N}(t,0) \end{pmatrix}$$

where for simplicity, we have dropped the limits of integration from eq. (45). Rows sum to the number of new infections occurring in neighbourhood j at time t. This formulation makes explicit the relative contributions of each neighbourhood c to the new human infections in neighbourhood j at time t. As before, the diagonal elements describe the within-neighbourhood transmission from the renewal equation.

We can define the aggregate number of new infections using an across-location renewal equation:

$$|E(t,0)| = \sum_{j=1}^{N} \sum_{c \in \mathcal{N}(k)} R^{cj}(t) \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} g^{cj}(t, a_E, a_I, a_W, a_V) \cdot E^c(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W da_V, \tag{46}$$

which is the grand sum of the renewal equation matrix.

To make explicit the row sum calculation that gives rise to the expected number of new infections occurring in human residents of neighbourhood j at time t, we write:

$$E^j(t,0) = \sum_{c \in \mathcal{N}(k)} R^{cj}(t) \int_0^t \int_0^{t-a_V} \int_0^{t-a_V-a_W} \int_0^{t-a_V-a_W-a_I} g^{cj}(t, a_E, a_I, a_W, a_V) \cdot E^c(t - a_V - a_W - a_I - a_E, 0) da_E da_I da_W \tag{47}$$

3.3 Resulting dynamics; GT, R(t), Incidence

3.4 The effects of epidemiological and human mobility influences on outbreak dynamics

- Effect of GT, Rt.
- Which features (e.g. length of stay, distance, number of trips) matter most? How varies by city?
- Which patches most influential? How varies by city
- High-degree nodes effects etc

- Effects: Time-varying + overall sensitivity (which vars most sensitive to, i.e. which assumptions most important to get right), overall epidemic size etc e.g. wrt

- within-neighbourhood time fraction and within-neighbur
- cross-neighbourhood fraction
- commuting duration
- long-distance flows

Repeat for system-level, inward- and outward-neighbourhood, pairwise $R(t)$ and incidence (peak, time-varying, aggregate)

3.5 Epidemic controllability: Counterfactual predictions

Reactivity of the system:

$$\sigma(t) = \mathcal{R}(t) \left(\frac{\mathbf{R}(t) + \mathbf{R}(t)^\top}{2} \right), \tag{48}$$

which is the largest eigenvalue of the symmetric part of $\mathbf{R}(t)$. $\sigma(t)$ can be greater than 1 even when $\mathcal{R}(t)$ is less than 1, meaning that perturbations can amplify before the eventual asymptotic decay. This will happen when we have asymmetric mobility patterns and hence, asymmetric $\mathbf{R}(t)$. This means we can capture transient dynamics and the effects of perturbations (e.g. interventions or importations)

Controllability: Can operate on restricted set of nodes.

Advantage is have causally explicit model that does not ignore external influences on a location's outbreak. Quantify:

- Elasticity: % reductions in R(t), peak, length of outbreak by different mobility interventions

4 Discussion

Introduce a framework that addresses a fundamental spatial limitation of most common way to estimate $R(t)$ as extend familiar renewal equation to allow for movement of humans (both infected and susceptible) across locations. This allows us to examine source-sink dynamics across mobility networks and capture how epidemics can be brought under control by targeted interventions. Our framework captures the wide array of within-day infection opportunities, created by the amount of time that individuals spend in their home location and other locations.

Discuss how to use pairwise reproduction numbers (reduce that specific between-location mobility flow temporarily), inward reproduction numbers (reduce inward flow), outward reproduction numbers (reduce outward flow). Generation time distributions also important not just for estimating $R(t)$ more reliably, also for knowing *when* and *how fast* infections will occur (not just magnitude) + estimating $r(t)$, *type* of intervention here. How long do we have to respond.

Previous studies: Type of human mobility data matters...type of human movement matters; distance, length of time, all of which important. Many studies have performed association studies where the effects of mobility on transmission are inferred/deduced using association between mobility and $R(t)$. However, this approach abstracts away the true causal mechanisms by which human movement shapes epidemics. In future studies, counterfactual predictions (a gold standard for causal inference) using different human movement patterns could be obtained using our causally explicit, mechanistic framework. This is not just a statistical issue, but also a trust issue for modelling and public health policy as inferred effects and information for decision-makers are not confounded by external factors.

All esp relevant for inference. High-quality or relevant mobility data/estimates. Shown biases of estimating independently. Unveiled the role of human movement. Practical guidelines, write down discrete form, which data to collect (sensitivity metrics)

Formulation of $f_i j$ allows for long-term/distance migration, while not explicitly considered.

within-day infection opportunities. Notes: Demonstrate t - Human mobility importance, policy implications - Type of human mobility data matters... - Similarly, type of human movement matters; distance, length of time, - Pathogen types/transmission routes -

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